

Quiz (Carolina Reaper)

Q1. The graph of $y = |x - 3|$ has a vertex at

- (A) (0, 3)
- (B) (3, 0)
- (C) (3, 3)
- (D) (-3, 0)
- (E) (0, 0)

Q2. The axis of symmetry for $y = |2x + 1|$ is

- (A) $x = -1$
- (B) $x = -0.5$
- (C) $x = 0$
- (D) $x = 0.5$
- (E) $x = 1$

Q3. The graph of $y = -|x - 2|$ opens

- (A) upward
- (B) downward
- (C) leftward
- (D) rightward
- (E) not at all

Q4. A shipping company uses the function $y = |x - 5|$ to model the distance between its trucks and a delivery point. If $x = 8$, the distance is

- (A) 3 miles
- (B) 5 miles
- (C) 8 miles
- (D) 10 miles
- (E) 13 miles

Q5. The graph of $y = |x + 2| - 3$ has a vertex at

- (A) (-2, -3)
- (B) (-2, 3)
- (C) (2, -3)
- (D) (2, 3)
- (E) (0, -3)

Q6. A logistics company uses the function $y = 2|x - 1|$ to model the cost of shipping packages. If $x = 4$, the cost is

- (A) 2
- (A) 4
- (A) 6
- (A) 8
- (A) 10

Q7. The graph of $y = |3x - 2|$ has an axis of symmetry at

- (A) $x =$

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- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$

Q8. The graph of $y = -2|x + 1| + 1$ opens

- (A) upward
- (B) downward
- (C) leftward
- (D) rightward
- (E) not at all

Open Ended Questions (FRQ)**Q9.** Graph the function $y = |x - 2| + 1$ and identify its vertex, axis of symmetry, and direction of opening. Show all work and explain your answers.**Q10.** A delivery company uses the function $y = |x - 3| - 2$ to model the time it takes to deliver packages. If $x = 5$, what is the time it takes to deliver the package? Show all work and explain your answer.

Chef's Tips

- Tip 1: Ensure students understand the properties of absolute value functions
- Tip 2: Encourage students to use graphing tools to visualize the functions
- Tip 3: Remind students to show all work and explain their answers clearly